

Depot

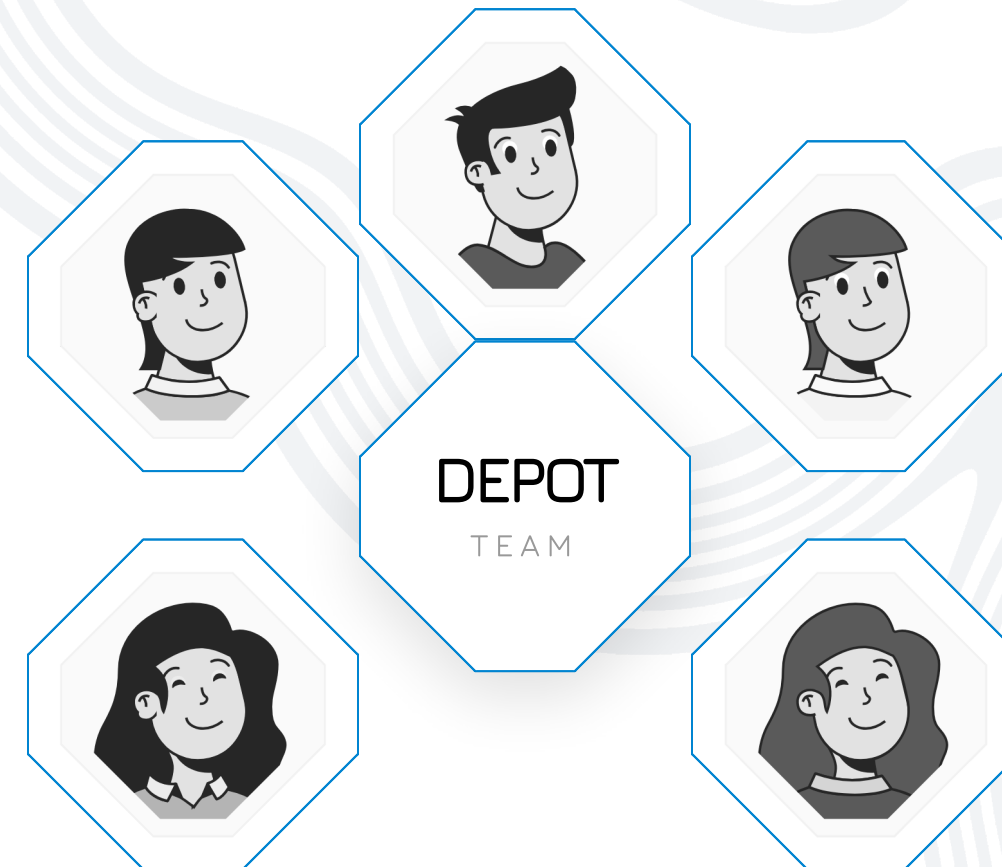
An infrastructure architect powered by AI,
from code to cloud.

TEAM

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Building apps is fast now. Shipping them is the bottleneck.

~70%

**of AI-written cloud
setups fail on their first
deploy**

Even from the best of six AI
models, tested on 153 real-world
scenarios

[T. Zhang et al., "Deployability-centric infrastructure-as-code
generation," ACM FSE, 2026](#) [1]

77%

**of AI-written cloud code
that works still fails a
security check**

Even from the best model tested

[F. Vargas et al., "Security-first evaluation of text-to-Terraform,"
arXiv:2608.02672, 2026](#) [2]

29%

**of company cloud spend
is wasted**

Estimated by 753 cloud decision-
makers

[Flexera, "2026 State of the Cloud Report," 2026](#) [3]

7 of 15

**developers found
choosing cloud services
or an architecture hard**

The hard step most often reported

Depot pilot survey of developers, Sep 2026 (15 respondents, mostly CS
students) [4]

Working code still leaves four decisions.

01

What it needs

Which servers, database and storage the app needs to run.

02

Where it runs

Which cloud company hosts it: Amazon, Google or Microsoft.

03

What it costs

The monthly bill, known before anything is switched on.

04

How to change it safely

Updating it later without breaking what already works.

Depot answers all four, **and explains, records and approves every change.**

Each tool solves part of it.

Someone still decides where it runs.

	Vercel	Porter	Pulumi Neo	Depot
Easy to deploy?	Yes Push code, it's live	Yes Connect a repository	No Needs infrastructure code	Yes Point it at your code
Runs on your own cloud?	No Vercel's platform only	Yes AWS, GCP or Azure	Yes Any cloud	Yes AWS, GCP or Azure
Cuts your cloud bill?	No Fixed pricing	No You watch the bill	No Not its focus	Yes Picks the cheapest fit, finds savings

From each product's own documentation [6]. Depot is proposed.

Built for builders without an infrastructure engineer.

AI-assisted builders

A working product, but little
experience of running one.

Solo full-stack developers

Outgrowing a managed
platform: stay, or move to a
cloud?

Small teams and agencies

Two to ten people, with no
DevOps staff.

What Depot does for you.

01

Reads your code

Works out what the app needs from its code and a few questions.

02

Picks the cloud

Compares AWS, Google Cloud and Azure, and chooses the best fit at the lowest cost.

03

Writes the infrastructure

Designs the architecture and writes the Terraform from proven, security-reviewed blueprints.

04

Checks and repairs its work

Validates, scans and prices every change, and fixes failures before you see them.

05

Ships and keeps it running

Deploys on every push, rolls back bad releases and tracks cost.

06

Fixes problems by chat

Diagnoses incidents and proposes fixes and savings for a person to approve.

Three objectives.

What a user can do with Depot.

01

Deploy any supported repository

Code in, live app out, on AWS, GCP or Azure.

02

Check health, metrics and cost

Logs, spend and rollback in one place.

03

Remembers every decision

What changed, when, and why.

Research sits inside all three: comparing ten current AI models on architecture and IaC writing, and measured guardrails for the agent loop.

One session, from working code to live.

Step 1

Connect

GitHub, GitLab or an archive

Point Depot at the app's code, or upload it.



Step 2

Answer

Up to eight questions

Answer simple questions the code can't, like expected traffic and budget.



Step 3

Review

AI-written architecture and Terraform, with monthly cost

See the proposed setup and what it will cost each month.



Step 4

Approve

Signed and role-checked

Nothing goes live until an authorised person signs it off.



Step 5

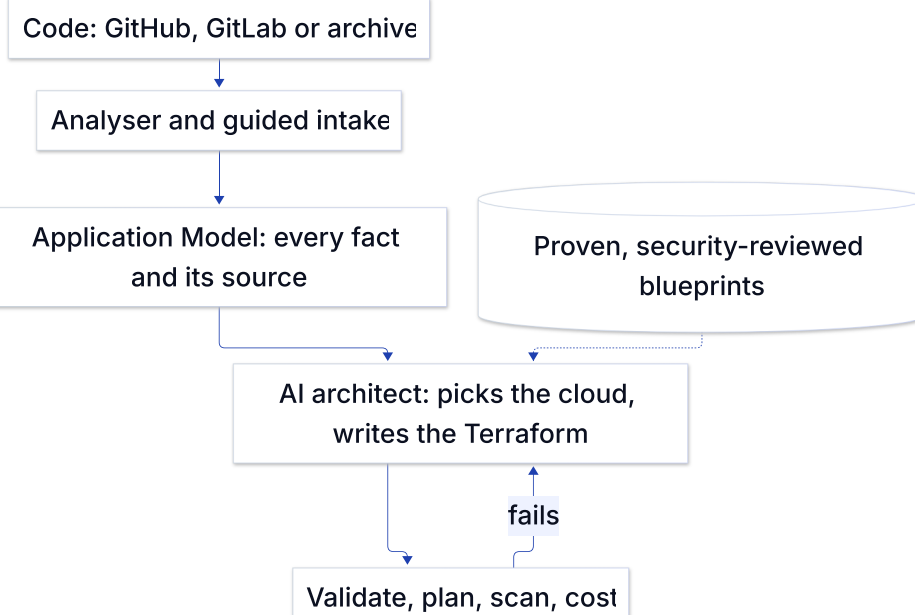
Run

Deploy on push, roll back, watch cost

The app goes live, updates itself, and can be undone.

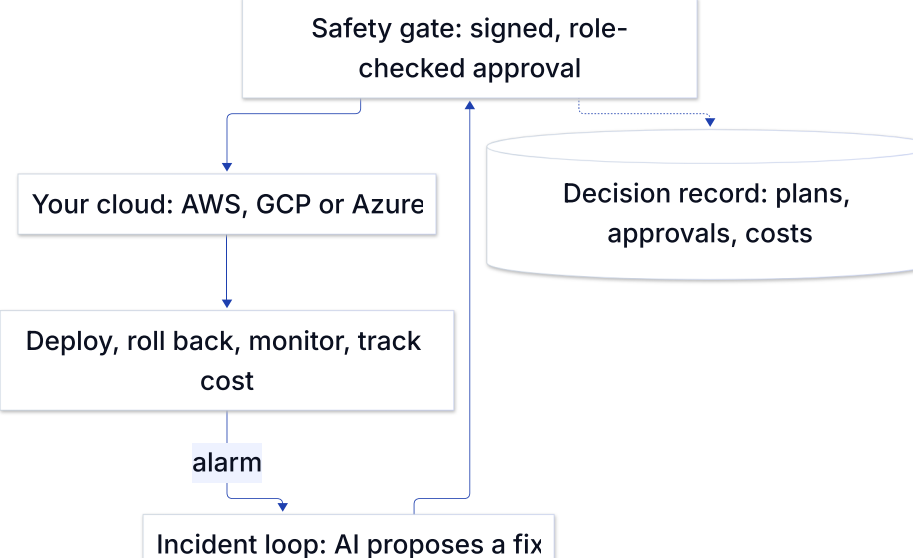
How Depot is built: from code to a running cloud.

1· Understand, design and verify



→
passes

2· Approve and run



One approval gate for every change, from the dashboard, the chat assistant or an incident fix.

What Depot covers.

**Any web app, API, static
site or worker**

Written in JavaScript/TypeScript,
Python or Go

**From GitHub, GitLab or
an upload**

One code source per app

**On AWS, Google Cloud or
Azure**

In the user's own cloud account

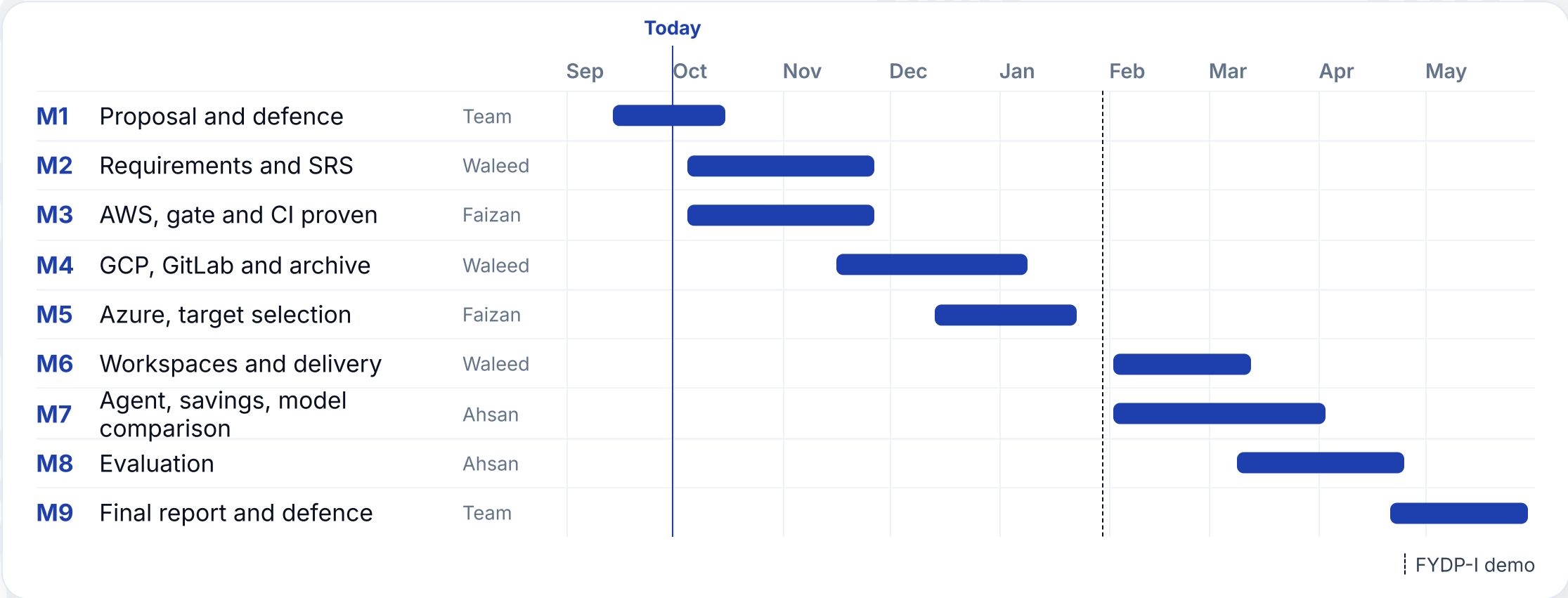
How we'll prove it works.

Pass marks set before we build.

We measure	Pass if	Context
Deploys successfully	7 of 8 test apps live after repair	Today's AI: 20.8–30.2% on the first try [1]
Picks the right architecture	80% or more accepted by reviewers	Two independent reviewers
Understands the app	85% or more of facts correct	Checked against labelled apps
Estimates the cost	Within 25% of the cloud's own calculator	Before anything is switched on
Cuts the cost	20% or more saved each month	On apps that sit idle
Blocks risky changes	Zero unapproved changes	Every seeded unsafe plan blocked
Easy to use	80% deploy unaided; usability 68 or more [8]	68 is the published average

Pass marks from the FYDP-I proposal, measured in milestone M8.

Three clouds first, then everything else.



Team planning targets; dates will follow the official defence schedule.

The team.

No one reviews only their own work.



Ahsan Riaz

AI & data logic

- Reads code into the Application Model
- Agent mode and incident loop
- Model research and optimisation



Muhammad Waleed

Front end & delivery

- Dashboard and team workspaces
- Deploys, rollback and monitoring
- User research and testing



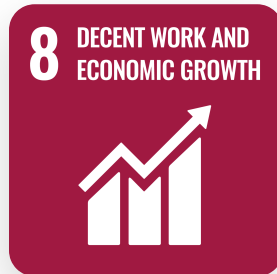
Muhammad Faizan Anwar

Cloud & backend logic

- AI architect and Terraform writing
- AWS, GCP and Azure adapters
- Safety gate and cost savings

Advisor **Hira Anwar** · Co-advisor **Ayesha Hakim**

Two goals we contribute to.



SDG 8.3

Small-enterprise growth

Small teams run products on cloud accounts they own, at a known monthly cost, without hiring.



SDG 9.5

Research and technological capability

An open benchmark, and views that explain why each architecture was chosen.

Icons are the official UN SDG icons, used under the UN SDG guidelines [7]. This publication has not been approved by the United Nations and does not reflect the views of the United Nations or its officials or Member States.

Sources.

- 1 [T. Zhang, S. Pan, Z. Zhang, Z. Xing and X. Sun, "Deployability-centric infrastructure-as-code generation: Fail, learn, refine, and succeed through LLM-empowered DevOps simulation," ACM FSE, 2026 \(arXiv:2506.05623\).](#)
- 2 [F. L. S. Vargas, R. B. Mansilha and D. Kreutz, "Security-first evaluation of text-to-Terraform: Benchmarking LLMs and SLMs for secure IaC generation," arXiv:2608.02672, 2026.](#)
- 3 [Flexera, 2026 State of the Cloud Report, 2026.](#)
- 4 Depot pilot survey of developers (15 respondents, mostly computer science students), NUST SEECS, September 2026.
- 5 Interview with the CEO of Lyte Studios, conducted with the same questions as the pilot survey, 2026.
- 6 Vendor documentation of the products compared.
- 7 [United Nations, "Goal 8" and "Goal 9," Sustainable Development Goals.](#)
- 8 [J. Sauro, "Measuring usability with the System Usability Scale \(SUS\)," MeasuringU, 2011.](#)

Click a source to open it. Full references are in the proposal.

Thank you. Any questions?

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